

Data Custody in LLM Alignment: Gradient Monitoring as an MLOps Security Gate

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Abstract. Preference-based alignment (RLHF, DPO) concentrates substantial behavioral influence in relatively small datasets, creating a security-critical supply chain in production MLOps pipelines. We present VSAT Monitor, a data-custody framework that treats preference datasets as production assets requiring validation before fine-tuning. The system performs a shadow gradient probe using a frozen checkpoint, compares reduced gradient representations against a versioned clean profile using covariance-aware scoring, and emits auditable alerts (allow, monitor, quarantine, block). We report an empirical characterization on two open-weight models—Pythia-70M-deduped and Qwen-2.5-1.5B—using five detection signals (Mahalanobis, ResidualPCA, spectral, cosine alignment, and loss shift) across three attack families at three poison rates. Our results indicate that gradient-based detection does not exceed chance level (AUROC ≈ 0.50) in centralized DPO, even with well-conditioned covariance estimation ($N_c = 5,377$, $d/N_c = 0.024$). Mahalanobis and ResidualPCA detect benign domain shift correctly but fail to separate coherent preference poisoning from clean gradient variance. These findings reinforce that production alignment pipelines require multi-layer data custody—provenance, statistical monitoring, and human review—rather than relying on post-hoc gradient detection alone.

Keywords: MLOps · LLMOps · data lineage · preference poisoning · direct preference optimization · gradient monitoring · DevSecOps · data custody · training observability · reproducible pipelines

1 Introduction

Preference-based post-training methods such as Direct Preference Optimization (DPO) [14] convert pairwise human judgments into high-leverage model updates. In production MLOps pipelines, preference datasets arrive from external sources: contracted annotators, third-party vendors, synthetic generation pipelines, or open collections [13]. Unlike pre-training corpora, preference data is a **security-critical production asset** that currently lacks standardized custody controls.

From an operational standpoint, the critical question before any fine-tuning run begins is:

Does this new preference batch induce training dynamics that are statistically incompatible with behavior historically observed for verified data from the same domain?

Existing MLOps data validation [13] and training observability [16] practices check schema, monitor distribution drift, and alert on loss anomalies. They do not treat preference data as a **chain of custody** with security gates before fine-tuning begins. Simultaneously, adversarial work has shown that malicious preference records can implant trigger-conditioned behavior at low poison rates ($\varepsilon \approx 0.5\%$) [12,15,3], and that existing spectral detectors from the vision literature fail to generalize to the LLM preference setting [12].

Contributions.

1. A **data-custody monitoring architecture** for DPO pipelines with immutable audit trails, four-state alert policy, and human-review gates.
2. **ResidualPCA**: a covariance-free detection signal that measures how much of a batch’s gradient falls outside the principal subspace of verified clean data, complementing Mahalanobis for coherent perturbations.
3. An **empirical falsification**: gradient-based monitoring does not separate poisoned from clean batches in centralized DPO on open-weight LLMs (Pythia-70M and Qwen-2.5-1.5B) at realistic poison rates. The failure is informative—it characterizes three distinct failure modes.

The monitor detects deviation, not malicious intent. A high score is evidence for human review, not proof of poisoning. This work does not claim a deployable defense; it presents a reproducible baseline and a set of falsifiable directions for the community.

2 Related Work and the MLOps Gap

2.1 Poisoning in Preference Alignment

Pathmanathan et al. [12] demonstrated at AAAI 2025 that preference poisoning is feasible and effective, implanting behavioral triggers at $\varepsilon = 0.05$ while preserving model utility. They also showed that spectral signature detection [17] and clustering-based filtering fail to generalize to the LLM preference setting. Rando and Tramèr [15] embedded universal jailbreaks via safety-preference manipulation. Baumgärtner et al. [3] showed in Best-of-Venom that RLHF can be attacked by injecting poisoned preference data to bias policy outputs toward attacker-chosen responses while maintaining plausible annotation disagreement. More recent work extends the threat to efficient offline attacks [18] and gradient-stabilization [11].

2.2 Data Validation and Training Observability

Polyzotis et al. [13] established data validation as a first-class MLOps concern covering schema conformity, distribution shift, and feature statistics. SageMaker Debugger [16] demonstrated real-time gradient monitoring as a training-observability tool. Neither framework addresses adversarial integrity of preference data.

2.3 Gradient-Space Defenses

Spectral Signatures [17] detect backdoor patterns in image classifiers by inspecting top singular values of per-class feature matrices. SPECTRE [6] extends this with robust covariance estimation for stronger guarantees. SEVER [4] applies robust statistics to gradient-based outlier removal. Influence functions [8] trace predictions back to training examples but are computationally prohibitive at LLM scale. None of these have been empirically validated for DPO on autoregressive LLMs with LoRA adapters.

2.4 Federated Defenses

ProtegoFed [19] is the first backdoor-free federated instruction-tuning framework, using frequency-domain gradient analysis to identify and remove interspersed poisoned samples across clients, achieving 92–100% detection with near-zero attack success rate. It operates at the optimization layer in a federated setting; VSAT Monitor operates at the data-custody layer in a centralized setting. The two approaches are complementary and we discuss their combination in Section 7.

2.5 The MLOps Gap

No prior work provides an empirical validation of gradient-based detection in centralized DPO on modern open-weight LLMs with LoRA adapters at realistic poison rates, with pre-registered hypotheses and domain-shift controls. This paper fills that gap.

3 Threat Model and Attack Taxonomy

We assume a **gray-box** attacker who can inject preference records before or during the fine-tuning data collection phase, without control over the optimizer, model weights, or deployment infrastructure. The poison budget is $\varepsilon = m/n$ (fraction of injected records). The joint success criterion requires simultaneous stealth and efficacy:

$$\text{ASR} \geq \tau_{\text{attack}} \wedge \Delta U \leq \tau_{\text{util}}, \quad (1)$$

where ASR is trigger-conditioned attack success rate and ΔU is utility degradation on clean held-out pairs.

Table 1. Attack taxonomy for preference-data poisoning in DPO pipelines.

ID	Attack	Mechanism	Detection Challenge
A1	Lexical Trigger	Insert rare token; invert preference label	Trigger absent from clean validation set
A2	Label-Flip	Reverse chosen/rejected target topic; no text change	Resembles annotation disagreement noise
A3	Output-Feature Injection	Append hidden trait to chosen response	phrase DPO loss indistinguishable from clean pairs

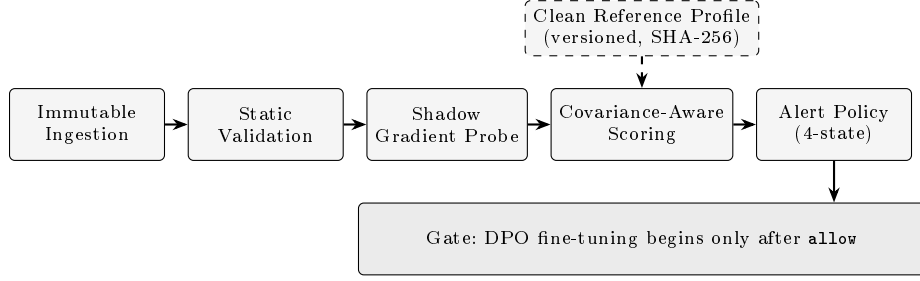


Fig. 1. VSAT Monitor pipeline. Each preference batch passes immutable ingestion, static validation, a frozen-checkpoint shadow gradient probe, and covariance-aware scoring against the versioned clean profile; the four-state alert policy gates the fine-tuning release.

Why A3 is the hardest case. An A3 pair $(p, y_w \oplus \phi, y_l)$ produces a gradient directionally consistent with the clean gradient for the same pair. The injected phrase ϕ contributes a small additive perturbation spread across the vocabulary embedding dimension, absorbed within clean gradient variance at $\varepsilon \leq 0.10$. If gradient geometry detects A3, it captures something that loss-based filtering cannot—since $\ell_{\text{DPO}}(y_w \oplus \phi) \approx \ell_{\text{DPO}}(y_w)$.

4 The VSAT Monitor Pipeline

Figure 1 summarizes the five-phase architecture.

4.1 Phase 1: Immutable Ingestion and Static Validation

Each incoming preference record (x, y^+, y^-) is stored with its content hash, annotator or source identifier, topic metadata, schema validation result, and dataset lineage in an append-only log. Static validation checks schema conformity, duplicates, label balance, topic distribution, and provenance coherence before any gradient computation occurs.

4.2 Phase 2: Shadow Gradient Probe

A frozen checkpoint θ_0 (trained on verified clean data) computes per-example gradients of the DPO loss with respect to a monitored parameter subset P_S (LoRA adapter weights) *without* performing a weight update:

$$g_i = P_S \nabla_{\theta_0} \ell_i^{\text{DPO}}(\theta_0). \quad (2)$$

Gradients are compressed via a Johnson–Lindenstrauss random projection [7,1]: $z_i = R(g_i) \in \mathbb{R}^d$ with $R \sim \mathcal{N}(0, 1/d)^{d \times |P_S|}$ (seed pre-registered and fixed).

4.3 Phase 3: Reference Profile Construction

Over N_{prof} verified-clean batches, collect N_c reduced gradient vectors $\{z_j\}$ and estimate the shrinkage covariance:

$$\hat{\Sigma}_\lambda = (1 - \lambda)\hat{\Sigma} + \lambda \tau I, \quad \tau = \frac{\text{tr}(\hat{\Sigma})}{d}, \quad (3)$$

where λ is the Ledoit–Wolf analytical coefficient [9]. The profile stores $(\hat{\mu}, \hat{\Sigma}_\lambda^{-1}, V_k, Q_{\text{ref}})$ plus all version identifiers. It is fingerprinted (SHA-256) and invalidated whenever any version identifier changes.

4.4 Phase 4: Detection Signals

Five signals are computed on each candidate batch $\mathcal{B} = \{z_i\}_{i=1}^B$:

S1 — Mahalanobis (batch percentile).

$$s_{\text{maha}}(\mathcal{B}) = Q_{0.90} \left\{ (z_i - \hat{\mu})^\top (\hat{\Sigma}_\lambda + \varepsilon I)^{-1} (z_i - \hat{\mu}) : i \in \mathcal{B} \right\}. \quad (4)$$

S2 — Spectral. Energy fraction of the first principal component of the centered batch matrix.

S3 — Cosine Alignment. Mean pairwise cosine similarity among the top-25% Mahalanobis outliers.

S4 — Loss Shift. Standardized batch mean DPO loss relative to the clean loss profile.

S5 — ResidualPCA. Let $V_k \in \mathbb{R}^{d \times k}$ be the principal subspace of the clean profile capturing fraction $\eta = 0.85$ of variance. For each $z_i \in \mathcal{B}$:

$$r_i = (z_i - \hat{\mu}) - V_k V_k^\top (z_i - \hat{\mu}). \quad (5)$$

The batch score aggregates individual residual norms:

$$s_{\text{RPCA}}(\mathcal{B}) = \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \|r_i\|_2 + \sigma(\{\|r_i\|_2\}_{i \in \mathcal{B}}), \quad (6)$$

Table 2. Alert policy. Thresholds calibrated on clean reference data and frozen before any poisoned evaluation. No state automatically modifies preference labels.

State	Evidence	Pipeline action
allow	All scores within reference limits	Continue; append decision to audit log
monitor	Isolated or moderate deviation; no persistence	Continue with enhanced logging; trigger sample review
quarantine	Elevated outlier fraction or coherent shift; or two consecutive monitor	Quarantine batch; require independent human review
block	Persistent severe deviation and provenance failure	Block release; open security incident; rebuild dataset manifest

where $\sigma(\cdot)$ is the sample standard deviation. This formulation (mean + std) is robust to single-example outliers in the batch. A high s_{RPCA} indicates gradient directions *new* relative to clean-data variance, even when per-example magnitudes are normal—the case targeted by A3 injection. This extends the conceptual approach of SPECTRE [6] from robust covariance estimation (image classifiers) to a projection-based residual score for LoRA gradient space.

4.5 Phase 5: Alert Policy

Every decision is replayable from an append-only audit record containing: dataset hash, profile fingerprint, checkpoint identifier, projection seed, covariance estimator, score summaries, alert state, and reviewer action. The four-state policy is designed to integrate as a security gate in the fine-tuning stage of a standard CI/CD training pipeline (DevSecOps): only batches in **allow** state proceed automatically to training release.

5 Empirical Characterization

5.1 Chain-of-Custody Protocol

All hyperparameters (Table 3) were pre-registered and frozen before inspecting poisoned evaluation results. Evaluation uses the Mann–Whitney U statistic [10,2] as the AUROC estimator with 1,000-replicate bootstrap CIs [5] (seed 0). Detection thresholds are calibrated to $\text{FPR} \in \{0.01, 0.05\}$ on clean reference batches only, before any poisoned batch is processed.

Experiment logs: <https://wandb.ai/jt-mancilla-mexico/vsat-monitor>.

5.2 Pythia-70M: Capacity Baseline

With $\approx 98\text{K}$ LoRA parameters and 400 DPO steps, the training loss converges to 0.044, indicating near-saturation of the adapter’s representational budget. All

Table 3. Pre-registered experimental configuration (all values fixed before evaluation).

Parameter	Pythia-70M (baseline)	Qwen-2.5-1.5B (main)
LoRA rank r / α	8 / 16	8 / 16
LoRA target modules	attention	attention
DPO steps / β	400 / 0.1	100 / 0.1
Batch size B (micro-batch)	32 (32)	64 (4)
Profile batches / samples N_c	200 / 6,214	100 / 5,377
JL proj. dim d / d/N_c	2,048 / 0.33	128 / 0.024
Eval batches per seed / seeds	12 / {0,1,2}	12 / {0,1,2}
Poison rates ε	0.5%–10%	1%–10%
Dataset / hardware	HH-RLHF (English) / Apple M3 24 GB	

evaluated signals yield AUROC ≈ 0.49 – 0.51 across all attack families and poison rates (Mahalanobis: 0.491, Spectral: 0.493, Cosine: 0.494, Loss Shift: 0.508).

Diagnosis. The poison gradient is a rank-1 perturbation relative to clean gradient variance at this scale. With $N_c = 6,214$ and $d/N_c = 0.33$, the profile is well-conditioned, ruling out estimation error as a confound. Capacity ($\approx 100\text{K}$ parameters) is insufficient to produce a detectable geometric trace.

5.3 Qwen-2.5-1.5B Pilot: Ill-Conditioned Profile ($N_c = 276$)

A pilot run with 20 profile batches of size 16 produced $N_c = 276$ samples with $d/N_c = 0.93$. Mahalanobis AUROCs for A1 ($\varepsilon = 0.05$) reached 0.531 with bootstrap CIs spanning $[0.19, 1.00]$ —statistically vacuous.

Diagnosis. With $d/N_c \approx 1$, Ledoit–Wolf shrinkage is insufficient to stabilize $\hat{\Sigma}$; the precision matrix amplifies noise in under-sampled directions. This is a co-variance estimation artifact, not a positive detection result. Resolution: increase N_{prof} to achieve $d/N_c < 0.03$ (requires $N_c \geq 4,000$ for $d = 128$).

5.4 Qwen-2.5-1.5B Main Run: Signal Collapses to Chance

With $N_c = 5,377$ and $d/N_c = 0.024$, Ledoit–Wolf operates in a well-conditioned regime. All five signals collapse to AUROC ≈ 0.50 across all conditions (Table 4, Figure 2). No condition exceeds 0.536, and no value is statistically distinguishable from chance at $p < 0.05$ by bootstrap test.

5.5 Control Analysis: Domain-Shift Confounding

We evaluate all signals on two benign conditions: (1) batches from the *medicine* domain (not seen during profiling), and (2) batches with random 10% label noise. Table 5 and Figure 3 report the results.

Key positive finding. Mahalanobis and ResidualPCA achieve AUROC = 0.00 on domain-shift: out-of-domain batches score lower than clean batches because OOD prompts produce smaller LoRA policy gradients under the domain-specific DPO objective. These two signals are **admissible** for operational use:

Table 4. AUROC on Qwen-2.5-1.5B main run ($N_c = 5,377$, $d = 128$, 3 seeds $\times$ 12 eval batches). **Bold**: best per attack at $\varepsilon = 0.10$. All values are consistent with chance (AUROC = 0.50).

Attack	ε	Maha	RPCA	Spec	Cos	Loss
A1 Lexical	0.01	0.507	0.502	0.495	0.503	0.503
A1 Lexical	0.05	0.532	0.520	0.512	0.484	0.505
A1 Lexical	0.10	0.536	0.518	0.490	0.462	0.497
A2 Label-Flip	0.01	0.503	0.499	0.496	0.507	0.509
A2 Label-Flip	0.05	0.508	0.509	0.515	0.508	0.503
A2 Label-Flip	0.10	0.516	0.520	0.508	0.520	0.499
A3 Output-Feature	0.01	0.501	0.507	0.500	0.502	0.499
A3 Output-Feature	0.05	0.501	0.515	0.496	0.506	0.513
A3 Output-Feature	0.10	0.497	0.519	0.498	0.493	0.501

Table 5. Control AUROC ($N_c = 5,377$). Ideal: Domain Shift $\rightarrow$ 0.00 (OOD batches should score lower than clean); Label Noise $\rightarrow$ 0.50. **Red** = inadmissible signal.

Signal	Domain Shift	Label Noise	Admissible
Mahalanobis	0.000	0.431	✓
ResidualPCA	0.000	0.424	✓
Spectral	0.007	0.465	✓
Loss Shift	0.021	0.417	✓
Cosine	1.000	0.559	✗

they correctly suppress false alarms on benign distribution shifts. Cosine alignment (AUROC = 1.00 on benign OOD) is **categorically inadmissible** as a standalone signal in multi-domain pipelines.

6 Discussion: Why Gradient Monitoring Fails

The covariance assumption is violated. Mahalanobis assumes the clean gradient distribution is approximately elliptical. DPO gradients over HH-RLHF are multi-modal: each topic, response length, and linguistic register produces a distinct cluster. Ledoit–Wolf shrinkage collapses this structure into one ellipsoid, making distance-from-mean an unreliable discriminator for subtle perturbations. ResidualPCA partially addresses this—by capturing 85% of clean variance in a basis rather than a single covariance—but is insufficient to characterize the full multi-modal structure from $N_c = 5,377$ samples at $d = 128$.

Coherent poison is not an outlier. Vision backdoor attacks produce gradients with anomalously high magnitude in a specific direction, creating a detectable outlier in covariance space. An A3 output-feature injection does not: the gradient back-propagates an update directionally consistent with clean preference

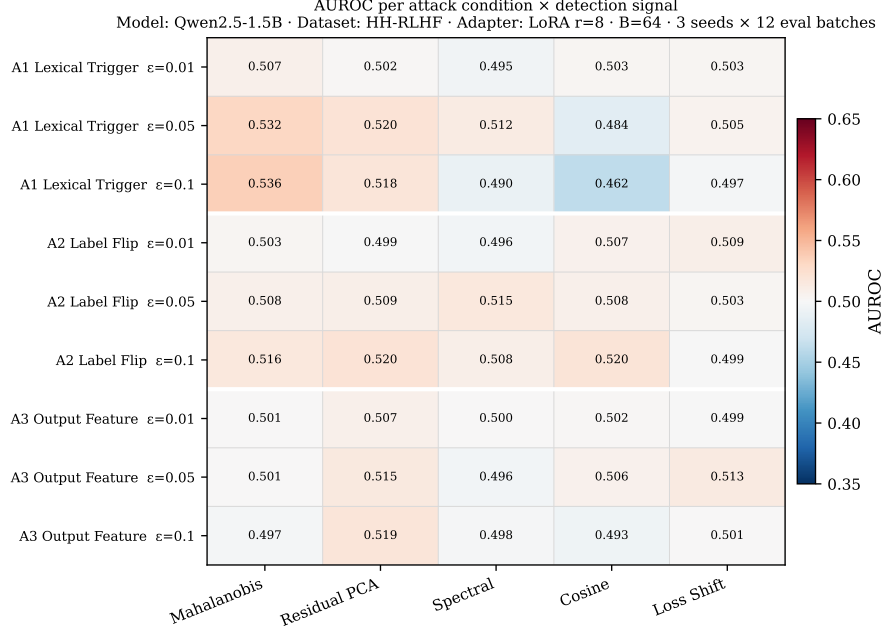


Fig. 2. AUROC heatmap: attack condition × detection signal (Qwen-2.5-1.5B, $N_c = 5,377$, $B = 64$, 3 seeds). All values at chance level (≈ 0.50). Hypothesis H1 (covariance-aware gradient monitoring detects preference poisoning in centralized DPO) is falsified.

gradients for the same pair. The injected phrase contributes a small additive perturbation spread across the vocabulary embedding dimension, completely absorbed within clean variance at $\varepsilon \leq 0.10$.

Scale dilutes the signal. At 1.5B scale and $d = 128$, the projected poison contribution per dimension is approximately:

$$\Delta \bar{z}_j \approx \frac{\varepsilon \cdot \Delta g_j}{\sqrt{d}} \approx 0.009 \sigma_g, \quad (7)$$

at $\varepsilon = 0.10$, $B = 64$, where σ_g is the clean standard deviation per projected dimension. With $n = 36$ paired observations per condition (3 seeds × 12 batches), this SNR is below detection threshold.

Implication for MLOps. These three failure modes converge on one operational conclusion: in centralized DPO, gradient geometry alone cannot serve as a firewall against subtle preference poisoning. This reinforces the need for multi-layer custody as a system property, not a single control.

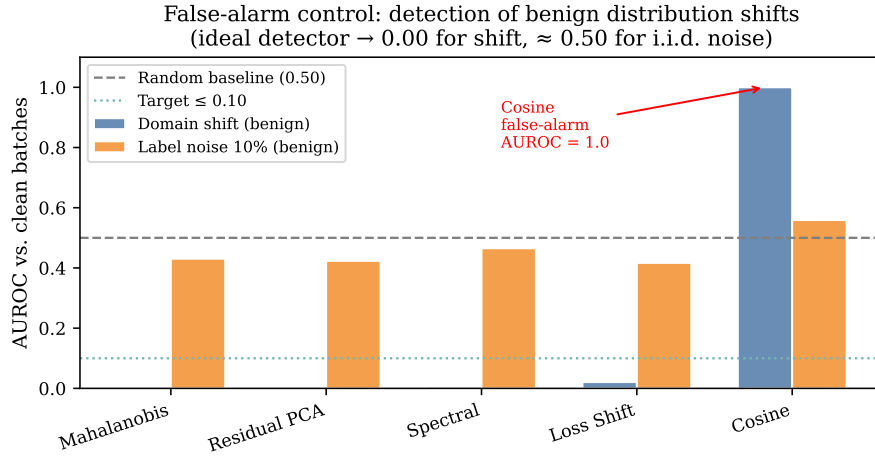


Fig. 3. Control AUROC on benign inputs. Cosine alignment scores 1.00 on domain-shift batches—confirming it is inadmissible in any multi-domain pipeline. Mahalanobis and ResidualPCA score OOD batches *lower* than clean (AUROC = 0.00), consistent with their geometric interpretation.

6.1 Research Questions Answered

- **RQ1 (Detectability):** No, at realistic poison rates on open-weight models (AUROC $\approx$ 0.50).
- **RQ2 (Comparative value):** Mahalanobis and ResidualPCA are comparably uninformative for poison, but more robust to false alarms than cosine.
- **RQ3 (Specificity):** Mahalanobis and ResidualPCA suppress domain-shift false alarms; cosine fails categorically.
- **RQ4 (Overhead):** the shadow probe costs $0.66\times$ a DPO training step per pair ($3,390\pm 307$ vs. $5,133\pm 698$ ms/pair, mean over 5 batches, Apple M3/MPS); scoring is negligible (< 5 ms/batch). Profile construction scales linearly with N_{prof} (one probe per clean batch); Ledoit–Wolf refits are negligible (≤ 0.03 s at $N_c = 10,754$).
- **RQ5 (Stability):** Profile is stable under frozen version tuple (model, tokenizer, projection seed, software stack).

7 Federated and Regional Extensions

7.1 Federated Custody

In a federated alignment setting, preference data originates from distributed annotators or institutions. The VSAT Monitor framework extends naturally: each client maintains a local clean profile, computes shadow-probe scores locally without sharing raw gradients, and contributes only aggregated score vectors to a

global aggregator. This preserves data privacy while enabling cross-silo integrity verification.

ProtegoFed [19] demonstrated near-perfect detection of interspersed back-door data in federated instruction tuning via frequency-domain gradient analysis. The key distinction is that ProtegoFed operates at the optimization layer (filtering during training), while VSAT Monitor operates at the data-custody layer (gating before training begins). A combined architecture that uses ProtegoFed-style frequency analysis within the VSAT shadow probe represents a direct and technically motivated next step.

7.2 Regional Corpus: Spanish-Language Cybersecurity

Latin American cybersecurity operations require preference datasets in Spanish with heightened provenance requirements. Existing open preference corpora (HH-RLHF, HelpSteer) are English-only. A curated Spanish-language preference corpus for security scenarios—with chain-of-custody tracking from annotation through fine-tuning—is proposed as a regional contribution aligned with BeMoSys’s Latin American focus.

8 Conclusion

We presented VSAT Monitor, a data-custody framework for preference alignment pipelines, and empirically characterized its limits on two open-weight models. Three findings are falsifiable, reproducible, and informative for the community:

1. **H1 falsified.** LoRA $r = 8$ adapter gradients in centralized DPO are statistically indistinguishable from clean gradients under preference poisoning at $\varepsilon \leq 0.10$, across all signals, attacks, and seeds (AUROC ≈ 0.50).
2. **H2 confirmed.** Mahalanobis and ResidualPCA achieve AUROC = 0.00 on benign domain-shift batches, establishing them as the only admissible gradient-space signals for operational multi-domain custody systems.
3. **Cosine inadmissible.** AUROC = 1.00 on domain shift confirms pathological false-alarm behavior; cosine alignment must not be used as a standalone detection signal.

The failure is informative: production alignment pipelines must treat preference data as a custody chain with provenance, statistical monitoring, and human review. Future work: (1) layer-specific ablations (final attention layers only); (2) frequency-domain signals inspired by ProtegoFed; (3) higher-rank adapters ($r \geq 32$) to amplify detectable signal; (4) federated extension of the custody layer.

Data Availability. Code and configurations: <https://github.com/jtmancilla/vsat-monitor>.

W&B logs: <https://wandb.ai/jt-mancilla-mexico/vsat-monitor>.

All experiments use public datasets (HH-RLHF) and open-weight models.

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